Story 3: Methods and visual choices

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Question, audience and scope

Do stricter gun laws help reduce firearm mortality? This story helps students and policy readers compare 50 states in 2024. It identifies the state-level association and visible exceptions. It does not estimate the effect of passing a particular law.

API retrieval and outcome

The code queries the CDC NCHS open-data API at https://data.cdc.gov/resource/489q-934x.json. It filters for 2024 Q4, 12 months ending with quarter, Firearm-related injury, and Age-adjusted. One successful response contains all 50 state rates. The archived request URL, retrieval time, response and metadata are in data/raw/. Rates are deaths per 100,000 and include suicides, homicides and other firearm injury deaths. The API dataset is designated provisional and may be revised. This annual period is distinct from the three-month annualized series.

Five-point law scale

GIFFORDS 2024 grades supply a reproducible ordering of state law strength. The first letter defines the five bins: F = 1 (most lax), D = 2 (lax), C = 3 (middle), B = 4 (strict), and A = 5 (strictest). Plus and minus variants remain in the saved grade field. This is an ordinal Likert-style classification for the assignment, rather than an opinion survey. Unequal bin sizes are retained. GIFFORDS is an advocacy organization and the index reflects its policy choices. Its published death-rate column is excluded from this analysis.

Why these heat maps

The law and mortality maps use the same labeled state tile layout. Equal tile sizes keep small states visible; the schematic positions do not depict physical area or precise adjacency. The law map uses five ordered colors and the mortality map uses a fixed 0-to-30 sequential scale with printed values. Both use Viridis colors. The mortality-band heat map uses sequential blues to encode within-group percentages on a fixed 0-to-100 scale. Every cell prints both the state count and percentage, with the group size beside the row. Black outlines and connected annotations emphasize 15 of 22 F-grade states (68%) at rates of 16 or more and 7 of 13 A-grade states (54%) below 8. Numbered outlines on the mortality map locate the extremes and New Mexico exception. Printed law levels support the law-map color scale. These on-chart cues strengthen saliency, while geographic and category-based heat maps offer variety without adding chart types outside the assignment. Rounded percentages may not sum to 100%. Separate legends prevent viewers from treating law levels and deaths as the same numerical quantity.

Analysis and limits

The A-grade group averages 8.4 deaths per 100,000 versus 18.6 among F-grade states. Spearman correlation between ordinal law level and mortality is -0.717. New Mexico (B-, 26.6) is an exception. Category means treat each state equally and do not equal a pooled national rate. The B-group mean exceeds the C-group mean, so the relationship is not a uniform gradient. Law timing, ownership, economic conditions, demographics, geography, enforcement and neighboring laws remain uncontrolled. The 2024 scorecard and mortality are contemporaneous. A causal study needs policy implementation dates, multiple years and suitable comparison groups.

Conclusion: states with stronger 2024 law grades tended to have lower firearm mortality in 2024. The evidence supports an inverse association while leaving causation unresolved.

Sources and software

CDC NCHS API: https://data.cdc.gov/resource/489q-934x.json
CDC definitions: https://www.cdc.gov/nchs/state-stats/deaths/firearms.html
GIFFORDS 2024 scorecard: https://giffords.org/resources/scorecard2024/
Retrieved September 27, 2026. R generated the heat maps, replacing the Python chart applications in Stories 1 and 2. Python handles source retrieval and tables. See the notebook and sources/SOURCES.md for complete checks.

All 50 state assignments and rates

GIFFORDS 2024 source grade, derived law level, and 2024 CDC annual age-adjusted firearm mortality per 100,000.

State	Source grade	Law level	Deaths / 100,000
Alabama	F	1	23.7
Alaska	F	1	24.4
Arizona	F	1	16.9
Arkansas	F	1	20.6
California	A	5	7.0
Colorado	A-	5	15.3
Connecticut	A	5	5.8
Delaware	A-	5	11.8
Florida	C-	3	12.7
Georgia	F	1	17.6
Hawaii	A-	5	3.7
Idaho	F	1	15.9
Illinois	A-	5	12.4
Indiana	D-	2	17.2
Iowa	F	1	11.8
Kansas	F	1	15.2
Kentucky	F	1	18.5
Louisiana	F	1	22.9
Maine	C+	3	11.9
Maryland	A-	5	11.6
Massachusetts	A	5	3.8
Michigan	B-	4	11.9
Minnesota	B	4	9.8
Mississippi	F	1	28.0
Missouri	F	1	19.6
Montana	F	1	20.2
Nebraska	C-	3	10.9

State	Source grade	Law level	Deaths / 100,000
Nevada	B-	4	17.2
New Hampshire	D-	2	10.2
New Jersey	A	5	4.0
New Mexico	B-	4	26.6
New York	A	5	4.4
North Carolina	C-	3	16.3
North Dakota	F	1	12.2
Ohio	D-	2	14.6
Oklahoma	F	1	19.1
Oregon	A-	5	14.2
Pennsylvania	B	4	11.9
Rhode Island	A-	5	4.6
South Carolina	F	1	19.5
South Dakota	F	1	17.3
Tennessee	F	1	19.8
Texas	F	1	13.9
Utah	F	1	13.3
Vermont	B-	4	10.6
Virginia	B+	4	12.8
Washington	A-	5	11.0
West Virginia	F	1	15.3
Wisconsin	C	3	12.1
Wyoming	F	1	23.4